

Hyunsu Go

Undergraduate Researcher in Representation Learning for Biomedical and Visual Data

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Education

Seoul National University, College of Engineering

Mar. 2021 – Present

B.S. Candidate in Mechanical Engineering; Double Major in Computer Science and Engineering and Economics.

GPA: 4.14/4.30; Total Credits: 180. Relevant coursework: Robot Vision; Mathematical Foundations of Deep Neural Networks; Deep Learning: Statistical Perspective; Concepts and Applications in Probability; Linear Algebra.

Seoul Science High School

Mar. 2018 – Feb. 2021

Research Interests

Representation learning for structured biomedical and visual data; medical image analysis; learning-based 3D vision and human-centric perception.

Publications

LoSA-Net: A Localized and Scale-Adaptive Network for Boundary-Sensitive Prediction of Perineural Invasion in 3D MRI

ISBI 2026

Youngung Han, **Hyunsu Go**, Kyeonghun Kim, Induk Um, Junga Kim, Jaewon Jung, Nam-Joon Kim, et al.
IEEE International Symposium on Biomedical Imaging. Accepted, Oral Presentation.

MMA-Former: Multi-Window Mixture-of-Head Attention Transformer for Adaptive PNI Prediction in 3D MRI

ISBI 2026

Youngung Han, Induk Um, Kyeonghun Kim, Junga Kim, **Hyunsu Go**, Jaewon Jung, Nam-Joon Kim, et al.
IEEE International Symposium on Biomedical Imaging. Accepted, Oral Presentation.

MAESIL: Masked Autoencoder for Enhanced Self-Supervised Medical Image Learning

ICEIC 2026

Kyeonghun Kim, Hyeonseok Jung, Youngung Han, Junsu Lim, YeonJu Jean, Seongbin Park, Eunseob Choi, **Hyunsu Go**, et al.
International Conference on Electronics, Information, and Communication. Accepted.

Manuscripts Under Review

MATHENA: Mamba-based Architectural Tooth Hierarchical Estimator and Holistic Evaluation Network for Anatomy

MICCAI 2026

Kyeonghun Kim, Jaehyung Park, Youngung Han, Anna Jung, Seongbin Park, Sumin Lee, Jiwon Yang, Jiyeon Han, **Hyunsu Go**, et al.

NEMESIS: Noise-Suppressed Efficient MAE with Enhanced Superpatch Integration Strategy

AICAS 2026

Kyeonghun Kim, Hyeonseok Jung, Youngung Han, **Hyunsu Go**, Eunseob Choi, Seongbin Park, Junsu Lim, et al.

Anatomy-Privileged Distillation with Token Routing for MRI-Based Prediction of Perineural Invasion

AICAS 2026

Hyunsu Go, Youngung Han, Kyeonghun Kim, Junga Kim, Dohyun Kweon, Jinyong Jun, Sungha Park, et al.

SpikeDS: Dual Sparsity Spikformer for Perineural Invasion Prediction in 3D MRI

AICAS 2026

Induk Um, Youngung Han, Kyeonghun Kim, Yulgi Jeong, Jina Jeong, **Hyunsu Go**, Dohyun Kweon, et al.

Adaptive Routing for Efficient Diffusion Transformer-Based PNI Prediction

AICAS 2026

Youngung Han, Dohyun Kweon, Kyeonghun Kim, **Hyunsu Go**, Jina Jeong, Suah Park, Induk Um, et al.

Research Experience

Precision Bioinstrumentation Laboratory, Seoul National University	Jul. 2024 – Present
Undergraduate Researcher, Dept. of Mechanical Engineering Advisor: Prof. Joon Ho Kang. SMR-based single-cell signal analysis, representation learning, cross-experiment correction, and interpretable modeling.	
Imaging-driven Medical SuperIntelligence Group, Seoul National University	Aug. 2025 – Apr. 2026
Student Researcher, Interdisciplinary Program in AI Semiconductor Engineering Advisor: Prof. Nam-Joon Kim. 3D MRI analysis for PNI prediction, self-supervised learning, and efficient volumetric medical imaging models.	
CLean Energy & Nanoheat Laboratory, Seoul National University	Feb. 2022 – Jan. 2024
Undergraduate Researcher, Dept. of Mechanical Engineering Advisor: Prof. Sangwook Park. Energy systems modeling and digital-twin analyses for decarbonization, EV integration, and renewable-hydrogen systems.	

Selected Projects

Deformation-Robust Optical Flow Pretraining	Sep. 2025 – Dec. 2025
Robot Vision, Seoul National University Course project with Jaeseung Kim. Designed deformation-aware optical flow pretraining using deformed cost-volume autoencoding for robustness to non-rigid motion.	
Information and Communication Technology Team Lead	May 2023 – Sep. 2023
Seoul National University Social Responsibility. Led an appropriate-technology project for rural agriculture in Laos, developing sensor-based and solar-powered smart chicken coops.	

Teaching Experience

Lead Tutor, Exploring Computing: Applications in Everyday Life	Mar. 2026 – Present
Seoul National University. Project-based computing, GUI/game programming, event-driven interaction, web-based data visualization, and automation.	
Teaching Assistant, Introduction to Mechanical System Design and Robot Programming	Mar. 2023 – Jun. 2023
Seoul National University. Engineering drawings, CAD, MATLAB, Python, C/C++, and middleware-based robotic system integration.	
Peer Tutor, Mathematics for Economics	Sep. 2022 – Dec. 2022
Seoul National University Linear algebra, multivariable calculus, differential equations, and introductory optimization.	
Undergraduate Course Assistant, Creative Engineering Design	Sep. 2022 – Dec. 2022
Seoul National University Design fundamentals, kinematics, fabrication, and competition-based engineering projects.	

Honors and Scholarships

Semiconductor Scholarship, Seoul National University	Aug. 2025 – Present
Presidential Science Scholarship	May 2023 – Present
Best Presentation Award, International Conference on Energy, Aquatech and Sustainability (ICEAS)	Aug. 2024
For “Smart Chicken Coops: Implementing Appropriate Technology for Rural Agriculture in Laos.”	
Top Prize, Energy New Industry Theme Research Program	Jan. 2023
SNU Energy New Industry Convergence University Project; digital-twin-based smart grid research.	
Education and Research Foundation Scholarship, SNU College of Engineering	Aug. 2022; Feb. 2023

Technical Skills

Programming and tools: Python, MATLAB, C/C++, Java.
Academic preparation: deep learning, robot vision, algorithms, probability, statistics, econometrics, mechanical systems.